

William C Francis

Data Scientist | Machine Learning, Agentic AI & Robotics
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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL EXPERIENCE

Data Scientist, Chubb | May 2024 - Present

- Develop and deploy GLM pricing models for personal and commercial insurance, owning factor selection, validation, and deployment.
- Build production XGBoost models for quote conversion, retention, and loss cost to support underwriting, pricing, and portfolio strategy.
- Build reusable modeling tools that support the workflow from raw data to deployed rating models.
- Develop agentic AI workflows for structured extraction and natural-language exploration of portfolio data.
- Deliver customer lifetime value estimates through a Streamlit application on Snowflake to support acquisition and retention decisions.

Graduate Research Assistant, xLAB, University of Pennsylvania | Dec 2021 - May 2023

- Built perception and localization for an autonomous racing go-kart: PSPNet segmentation and GNSS/IMU fusion with an extended Kalman filter for 100 Hz odometry, deployed on NVIDIA Jetson AGX with TensorRT.
- Led the team to first place at Autonomous evGrandPrix 2023; presented the project at ICRA 2022.

EDUCATION AND HONORS

University of Pennsylvania | Dual M.S.E. in Robotics and Computer Science | May 2024

GPA: 3.8/4.0. Teaching Assistant, Applied Machine Learning. Coursework included deep learning, natural language processing, software systems, advanced computer vision, statistics, graph neural networks, and neuro-symbolic AI.

Vellore Institute of Technology | B.Tech. in Electrical and Electronics Engineering,

Minor in Computer Science | June 2020

Dean's List; VITEEE Scholarship awarded for the top 2% of the program.

TECHNICAL SKILLS

ML & AI: Python, C++, SQL, PyTorch, scikit-learn, XGBoost, GLMs, Hugging Face, prompt engineering, agentic workflows, structured extraction, LLM evaluation, Bayesian methods, time series, A/B testing.

Data & MLOps: Databricks, Snowflake, Spark, AWS, MLflow, Docker, Git, CI/CD, model monitoring, Agile.

ENTREPRENEURIAL INITIATIVES

PeeqAI | Founder | Dec 2023 | [Demo](#)

Built a hands-free assistive system for visually impaired users, combining a custom vision-language model, wake-word speech recognition, and GPT-4. Added a context layer for reasoning over accumulated environmental state.

Philly AI Connect | Co-Founder | Nov 2023 - May 2024 | [Website](#)

Grew the community to 1,800 members and organized six major events with talks, demos, workshops, and hackathons. Secured partnerships with Pennovation Center and GraysonSky.

PUBLICATION

Brain-Computer Interfacing for Wheelchair Control by Detecting Voluntary Eye Blinks

Indonesian Journal of Electrical Engineering and Informatics | June 2021

Developed neural-network blink detection and adaptive control for an EEG-driven wheelchair with camera-based collision avoidance.